A Blockchain-Enabled Framework for Misinformation Monitoring

Multilingual AI triage and a shared, tamper-evident ledger for South Africa's election fact-checkers

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Data Science for Social Impact

1. Introduction

In South Africa's 2024 election, AI deepfakes spread through WhatsApp groups and fact-checks were dismissed as propaganda [1]. Fact-checkers keep separate records, and detection tools rarely cover African languages:

67%
of detection studies use English data [2]

<10%
target low-resource languages [2]

RESEARCH QUESTION

How can an integrated AI and blockchain framework enable tamper-resistant, transparent, multilingual monitoring and verification of election misinformation in South Africa?

Aim: AI triage in three languages plus joint verdicts from independent fact-checkers on a shared, tamper-evident ledger.

2. Methodology

Design-science research: build the framework, then evaluate it against TF-IDF baselines over three seeds.

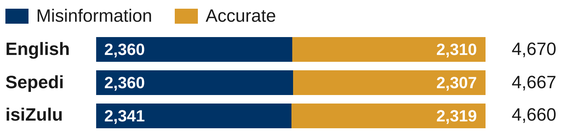
2.1 A four-stage pipeline

1. Ingest Kafka stream absorbs surges → **2. Triage** AfroXLMR [3] + LoRA [4] labels claims → **3. Verify** other orgs vote; majority finalises → **4. Record** hash on Fabric [5], text in IPFS

2.2 A synthetic multilingual corpus

No Sepedi or isiZulu election corpus existed:

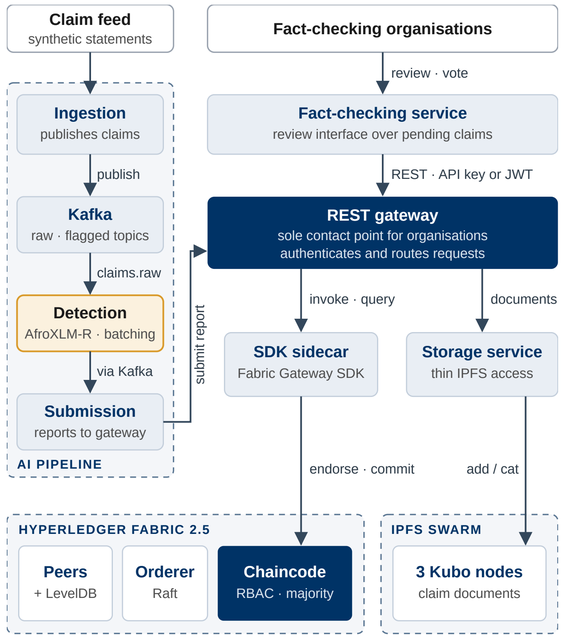
Prompts 5,000, fixed seed → **Generate** Gemini 3.6 Flash, English → **Translate** to Sepedi and isiZulu → **Clean, split** 13,877 kept, 80/10/10



All 13,997 generated statements, by language and label, before cleaning.

3. System architecture

A permissioned **Hyperledger Fabric** consortium [5] with Raft ordering: no fees, deterministic finality, and roles enforced inside chaincode from X.509 certificate attributes.



Claims flow through Kafka and the AfroXLM-R detection service to the REST gateway, the organisations' only point of contact. Votes go on-chain; claim text stays in IPFS.

3.1 Roles enforced in chaincode

Official submits, expires, admits members

Fact-checker submits reports and fact-checks

Observer read-only audit

3.2 POPIA by design

Minimality (s 10) Only hash, CID, AI metadata and org ID on-chain.

Purpose (s 13) Fixed schema; no free-form personal fields.

Retention (s 14) Claim text off-chain in IPFS, so deletable.

Security (s 19) SHA-256, TLS channels and chaincode RBAC.

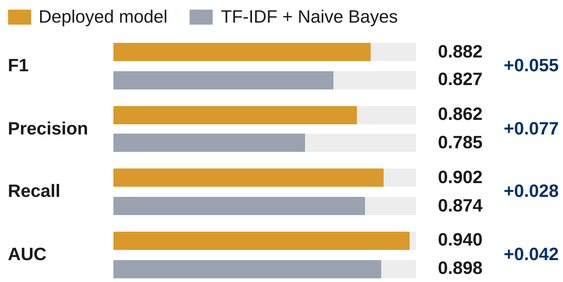
4. Performance of deployed model

Deployed detector: **AfroXLMR-large-76L** [3] with **LoRA** [4]. Means of three seeds, 1,393 test statements.

0.882 **F1** on the misinformation class, ± 0.015 across seeds; AUC 0.940

9 in 10 false claims are flagged for fact-checkers (recall 0.902), and **86%** of flagged claims really are false (precision 0.862).

4.1 Ahead of the strongest baseline



Bars start at 0.50; right column is the gain. The largest gain is precision, which means fewer false alarms for reviewers.

4.2 Why LoRA, not full fine-tuning

98% of full fine-tuning's F1 (0.882 vs 0.899)

1.29% of weights trained: a 7.3 M adapter on a shared base

≤ 0.04 loss gap at stop; 2 of 3 full runs overfit (0.40–0.49)

Full fine-tuning overfits synthetic data; LoRA should generalise better.

4.3 Holds up in Sepedi and isiZulu



Deployed model, F1 by language on the same axis as 4.1. The two African languages trail English by only about 0.04.

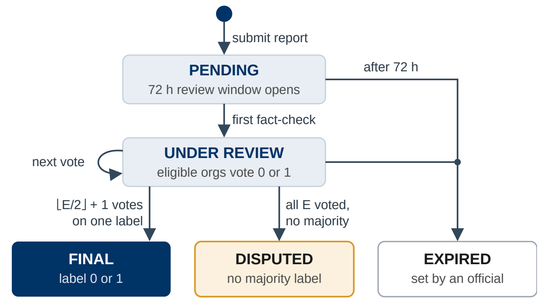
WHAT THIS MEANS FOR THE FRAMEWORK

The model triages; it does not decide. Its label and confidence are stored on-chain beside every vote, and the consortium (section 5) makes the final call, so model errors stay visible and can be outvoted.

4.4 Confusion matrix, mean of 3 seeds

	Flagged as false	Passed as accurate
False claim 704 in test set	**635** caught	**69** missed
Accurate 689 in test set	**102** false alarms	**587** cleared

5. Report lifecycle



E = organisations other than the submitter. Bottom row: final states.

6. Key takeaways

- **Novel design:** local-language AI, joint verdicts, shared ledger.
- **First Sepedi, isiZulu election detector:** near-English F1.
- **AI flags, people decide:** a majority of peer orgs sets verdicts.
- **Verified end to end:** all test workflows passed hash checks.
- **POPIA by design:** only hashes on-chain; claim text erasable.

Trained and tested on synthetic data; next: real election posts.

7. References

[1] G. Gondwe, "AI-driven disinformation and political influence on WhatsApp in South Africa's 2024 elections," *Int. J. Press/Politics*, 2025.

[2] S. Rananga, B. Isong, A. Modupe and V. Marivate, "Misinformation detection: a review for high and low-resource languages," *J. Inf. Syst. Informatics*, vol. 6, no. 4, 2024.

[3] J. O. Alabi, D. I. Adelani, M. Mosbach and D. Klakow, "Adapting pre-trained language models to African languages via multilingual adaptive fine-tuning," *COLING*, 2022.

[4] E. Hu et al., "LoRA: Low-rank adaptation of large language models," *ICLR*, 2022.

[5] E. Androulaki et al., "Hyperledger Fabric: a distributed operating system for permissioned blockchains," *EuroSys*, 2018.

Code and data (QR code, top right): github.com/SnowWoofer/A_Blockchain_Enabled_Framework_for_Misinformation_Monitoring